

Homework 5

Oct 15, 2025

1 (30'). The input is two binary numbers, x and y (with the most significant bit on the left), separated by a symbol $\#$, for example, $100\#100\sqcup$. (Assume there are no leading zeros unless the number is zero.) Accept if x and y are equal.

Draw the state diagram of the corresponding Turing machine and submit its source code on eLearning.

2. (30') Let language

$$L = \{0^n : n \text{ is a Fibonacci number}\}.$$

Design a Turing machine that decides L . Draw the state diagram of the corresponding Turing machine and submit its source code on eLearning.

3. (40') In the definition of the Turing machine, the head can move left or right. A Turing machine whose head can move left, right, or stay in place is called a *Turing machine with a stationary move*, or more formally:

$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}.$$

Prove that if a language L can be decided by a Turing machine with stationary moves, then L can also be decided by a standard Turing machine.